

Entropy generation on the variable electric field and EMHD SWCNT-blood nanofluid with melting/non-melting heat transfer

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Abstract

This work is concerned with the study of laminar, steady electro-magnetohydrodynamic flow and entropy generation of SWCNT-blood nanofluid with melting/non-melting heat transfer. A study is also underway to analyze the heat transfer phenomenon under factors such as viscous dissipation and joule heating. The homotopy perturbation method is employed to solve the coupled non-linear ordinary differential equations. The calculated results are plotted graphically through velocity, temperature, entropy generation, local skin friction coefficient, and rate of heat transfer. Melting/non-melting heat transfer is used in biomedical for example vanquish uses radio frequency to dangerously destroy fat cells. This model may use heat to melt away fat cells in the body. The entropy generation increases with increasing the electric field parameter. The heat transfer rate enhances with the higher values of the melting parameter.

Keywords

SWCNT-blood fluid, MHD, variable electric field, entropy generation and melting/non-melting heat transfer

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Introduction

Nowadays, many researchers and scientists are motivated by the heat transfer accompanied by a melting phenomenon due to its wide range of applications in mechanical and technological processes. In such a way melting phenomenon in the fluid flow of stretching surface is major vital in magma solidification, frozen ground thawing process, preparation of semiconductor material, welding and casting of a manufacturing process, melting of soil, melting of permafrost and crystal growth and so on.^{1,2} In addition, phase change materials (PCMs) have some profitable physical characteristics such as stability of chemical and thermal cycling that have allowed inclusive use. Because of these advantages, it has much potential for use in isolated thermal systems that rely on PCMs for thermal control and thermal energy storage. Epstein and Cho³ proposed that the melting heat mechanism was incorporated to examine the fluid flow properties over a flat plate. The outcomes show that the melting inhibits thermal transmission and decreases the number of Nusselt locally. Mabood⁴ numerically investigated the effects of melting and nonlinear thermal radiation on hybrid nanofluid ($Cu-Al_2O_3$ /water) over a stretching surface. The result shows that the fluid motion increments for enhancing melting parameter. Hayat et al.⁵ studied the influence of melting heat transfer in the boundary layer nanofluid flow determined for both effective Prandtl number and Prandtl number.

The flow transport phenomenon in porous media has abundant uses in environmental science, industrial and including hydrology such as catalytic reactors, soil physics (agriculture), electrochemical fuel cell devices, petroleum technology, mining and mineral processing, catalytic reactors, redesigned recovery of oil stores and warmth pipe development and heat removal from nuclear fuel debris, and so on.^{6–8} In the year 1856, a French civil engineer, Henry Darcy, laid the framework for the movement of homogeneous fluids across porous media during his work on the flow of water through sand beds. Darcy law cannot operate properly when higher flow rates of inertial and boundary effects occur. After that, in the year 1901, Dutch scientist Forchheimer⁹ extended the Henry Darcy velocity expression by joining the square velocity term in the momentum equation to predict the action of inertia and boundary layer flow. This factor is always efficient for a high Reynolds number. Muskar¹⁰ called this factor

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as “Forchheimer term.” Shah et al.¹¹ examined the Darcy-Forchheimer flow of the electromagnetic ferrofluid on a porous stretching/shrinking sheet under the effects of thermal radiations, external electric and magnetic fields. Eid and Mabood¹² investigated the Darcy-Forchheimer flow of MHD a micropolar dusty nanofluid over a permeable extending surface by taking into account the heat source/sink and thermal radiation. Kumar et al.¹³ studied the three-dimensional Darcy-Forchheimer stagnation point flow of a nanofluid over a stretching surface.

The investigation of the magnetic and electric fields has numerous applications in the mechanical, science, and medical areas, such as geothermal energy extraction to nuclear reactor coolers, receiving and transmitting antennas, electromagnetic pumps, fusion machines, magnetic resonance imaging equipment, cure of skin disorders, radiofrequency ablation used in cardiology, treatment of malaria infection, blood pumping, drug delivery, cancer therapy, and so on.^{14–17} Jakeer and Reddy¹⁸ discussed the electro-magneto-hydrodynamic significance on the stagnation-point flow of copper-silver/water hybrid nanofluid past an extending sheet. The outcomes exhibit that the rate of heat transfer is first enhancing and then diminishing to the greater values of the electric field. Reddy¹⁹ investigated numerically the steady 2D electrical magneto hydrodynamic (EMHD) stagnation-point flow of conducting hybrid nanofluid with non-uniform heat source/sink and nonlinear thermal radiation effects over a stretching sheet. Daniel et al.²⁰ examined unsteady 2D mixed convection flow of EMHD in the impact of slip and double stratification.

The following factors contribute to the authors of the study’s novelty. The study looks at nanofluid flow using Casson fluid as the base fluid and SWCNT nanoparticles floating in it. Non-zero potential flow has been used to account for the pressure gradient. The transverse magnetic field is a broader way of simulating magnetic lines of force acting at a distance. Because of the porous matrix, the flow is subjected to an extra body force. In the presence of a heat source, the electrically conducting fluid approaching the surface constantly melts into a warm liquid. In an electro-magnetohydrodynamic nanofluid, the flow is driven by the melting or non-melting surface and the initial conditions. Melting/non-melting heat transfer having several applications such as, magma solidification, melting of soil, melting of permafrost, crystal growth, to destroy tissues and cells, biopreservation, an innovative technology applied to tissue banking and cell banking, frozen ground thawing process, preparation of semiconductor material, plastic surgery treatment, assisted reproductive technologies (ART), welding and casting of a manufacturing process, cell therapeutics, tissue engineering, and organ transplantation.^{21–23}

Mathematical formulation

Here we have a keen attention to explore electro-magnetohydrodynamic SWCNT-blood nanofluid past over

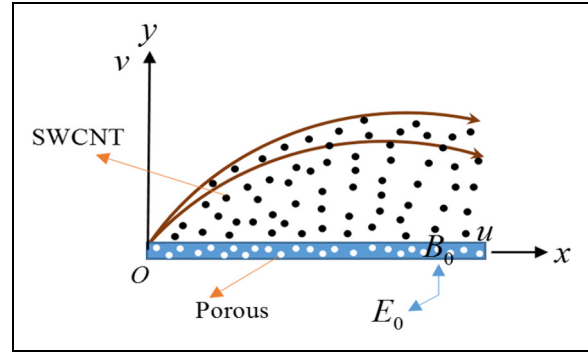


Figure 1. Geometry of the problem.

a melting/non-melting heat transfer surface. The melting/non-melting heat transfer sheet is affected with uniform magnetic field B_0 along y -direction (see Figure 1). In this model, the characteristics of viscous dissipation and electro-conductive heating are incorporated in the energy equation. The origin is fixed as a constant, and the sheet moving velocity is $u_w = U_0x$, where $U_0 > 0$ is the surface extending rate. It is assumed that the melting surface temperature is T_m and the temperature of the ambient fluid T_∞ is constant. Based on the above assumptions, SWCNT-blood nanofluid boundary-layer flow equations can be written as:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad (1)$$

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = v_{nf} \left(1 + \frac{1}{\beta} \right) \frac{\partial^2 u}{\partial y^2} + \frac{\sigma_{enf} B_0}{\rho_{nf}} (E_0 - B_0 u) - \frac{v_{hnf}}{k'} \left(1 + \frac{1}{\beta} \right) u - \frac{F^*}{k'} u^2, \quad (2)$$

$$u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \alpha_{nf} \frac{\partial^2 T}{\partial y^2} + \frac{1}{(\rho c_p)_{nf}} \frac{16\sigma^* T_\infty^3}{3k^*} \frac{\partial^2 T}{\partial y^2} + \frac{\mu_{nf}}{(\rho c_p)_{nf}} \left(1 + \frac{1}{\beta} \right) \left(\frac{\partial u}{\partial y} \right)^2 + \frac{\sigma_{enf}}{(\rho c_p)_{nf}} \{ B_0 u - E_0 \}^2 + \frac{Q_0}{(\rho c_p)_{nf}} (T - T_m) \quad (3)$$

The boundary conditions are

$$u = u_w, \quad T = T_m, \quad \text{at } y = 0, \\ u \rightarrow 0, \quad T \rightarrow T_\infty \quad \text{as } y \rightarrow \infty, \quad (4)$$

and

$$k_{nf} \left(\frac{\partial T}{\partial y} \right)_{y=0} = \rho_{nf} (\lambda + c_s [T_m - T_s]) v(x, 0). \quad (5)$$

The electrical conductivity σ_{enf} is assumed to be dependent on the velocity of the fluid and has the form

$$\sigma_{enf} = \sigma_{nf} u \quad (6)$$

where u and v are the x and y velocity components, respectively. F^* is the dimensional Forchheimer constant, c_s is the heat capacity of the solid surface, λ is the latent heat of the fluid, k^* is the mean absorption coefficient, k' is the permeability of the porous medium, T is the temperature of the fluid, T_s is the temperature of the solid surface, Q^* is the heat generation/absorption coefficient, and σ^* is the Stefan–Boltzmann constant.

The thermophysical properties and quantities of hybrid nanofluid are defined as

$$\begin{aligned} \mu_{nf} &= \frac{\mu_f}{(1-\phi)^{2.5}}, \nu_{nf} = \frac{\mu_{nf}}{\rho_{nf}}, \frac{\rho_{nf}}{\rho_f} = (1-\phi) + \phi \frac{\rho_{CNT}}{\rho_f}, \\ \frac{k_{nf}}{k_f} &= \frac{(1-\phi) + 2\phi \frac{k_{CNT}}{k_f} \ln \frac{k_{CNT} + k_f}{2k_f}}{(1-\phi) + 2\phi \frac{k_f}{k_{CNT} - k_f} \ln \frac{k_{CNT} + k_f}{2k_f}}, \\ \frac{(\rho c_p)_{nf}}{(\rho c_p)_f} &= 1 - \phi + \phi \frac{(\rho c_p)_{CNT}}{(\rho c_p)_f}, \\ \alpha_{nf} &= \frac{k_{nf}}{(\rho c_p)_{nf}}, \frac{\sigma_{nf}}{\sigma_f} = 1 \\ &+ \frac{3\phi \left(\frac{\sigma_{CNT}}{\sigma_f} - 1 \right)}{\left(\frac{\sigma_{CNT}}{\sigma_f} + 2 \right) - \phi \left(\frac{\sigma_{CNT}}{\sigma_f} - 1 \right)} \end{aligned} \quad (7)$$

where μ_{nf} is the viscosity of nanofluid, ϕ is the nanoparticle volume fraction, ρ_f and ρ_{CNT} are the densities of the fluid and solid particles, respectively, σ_f and σ_{CNT} are the electrical conductivities of the fluid and solid particles, respectively, and k_f , k_{CNT} are the thermal conductivities of fluid and nanoparticles, respectively.

The self-similar variables used to convert to ODE form are expressed as follows:

$$\begin{aligned} \eta &= \left(\frac{U_0}{v_f} \right)^{0.5} y, \quad \psi = (v_f U_0)^{0.5} x f(\eta), \\ \theta(\eta) &= \frac{T - T_m}{T_\infty - T_m}. \end{aligned} \quad (8)$$

Here ψ is the stream function and η is the similarity variable. The velocity components are $u = \frac{\partial \psi}{\partial y}$ and $v = -\frac{\partial \psi}{\partial x}$.

$$u = U_0 x f'(\eta), \quad v = -(v_f U_0)^{0.5} f(\eta) \quad (9)$$

where the prime denotes the derivative with respect to η . Clearly the continuity equation (equation (1)) is satisfied by applying the similarity transformations (equations (8) and (9)). By using equations (8) and (9), equations (2)–(7) are reduced as follows:

$$\begin{aligned} \frac{\mu_{nf}}{\mu_f} \left(1 + \frac{1}{\beta} \right) f''' - \frac{\rho_{nf}}{\rho_f} [f'^2 - f''f + F_s f'^2] \\ - \frac{\sigma_{nf}}{\sigma_f} M f' [f' - E] - K \frac{\mu_{nf}}{\mu_f} \left(1 + \frac{1}{\beta} \right) f' = 0, \end{aligned} \quad (10)$$

$$\begin{aligned} \frac{1}{Pr} \left(\frac{k_{nf}}{k_f} + \frac{4}{3} R \right) \theta'' + \frac{(\rho c_p)_{nf}}{(\rho c_p)_f} f \theta' + Ec \frac{\mu_{nf}}{\mu_f} \left(1 + \frac{1}{\beta} \right) f'^2 \\ + \frac{\sigma_{nf}}{\sigma_f} M Ec f' [f' - E]^2 + Q \theta = 0, \end{aligned} \quad (11)$$

with boundary conditions

$$\left. \begin{aligned} f'(0) &= 1, \quad \Pr \frac{\rho_{nf}}{\rho_f} f(0) + \frac{k_{nf}}{k_f} Me \theta'(0) = 0, \quad \theta(0) = 0, \\ f'(\infty) &= 0, \quad \theta(\infty) = 1. \end{aligned} \right\} \quad (12)$$

where K is the porosity parameter, Me is the melting parameter, E is the electric field parameter, Ec is the Eckert number, β is the Casson fluid parameter, F_s is the inertia coefficient, R radiation parameter, Pr is the Prandtl number, Q is the heat source parameter and M is the magnetic parameter. Which are given by

$$\begin{aligned} M &= \frac{\sigma_f B_0^2 x}{\rho_f}, \quad K = \frac{\nu_f}{k'_0 U_0}, \quad F_s = \frac{u_w F^*}{U_0 k'_0}, \quad \Pr = \frac{\mu_f (c_p)_f}{k_f}, \\ Ec &= \frac{u_w^2}{(c_p)_f (T_\infty - T_m)}, \quad R = \frac{4\sigma^* T_\infty^3}{3k^* k_f}, \\ Q &= \frac{Q_0}{U_0 (\rho c_p)_f} \text{ and } Me = \frac{(c_p)_f (T_\infty - T_m)}{(\lambda + c_s [T_m - T_s])}. \end{aligned}$$

Near the wall ($\eta = 0$), skin friction factor and rate of heat transfer are

$$\left. \begin{aligned} C_f \text{Re}_x^{1/2} / 2 &= \frac{1}{(1-\phi)^{2.5}} \left(1 + \frac{1}{\beta} \right) f''(0), \\ \text{Nu}_x \text{Re}_x^{-1/2} &= - \left(\frac{k_{nf}}{k_f} + \frac{4}{3} R \right) \theta'(0), \end{aligned} \right\} \quad (13)$$

Entropy modeling

The local volumetric rate of entropy generation in the presence of electromagnetic field and porous media is

given by equation (21)

$$S_G = \underbrace{\frac{1}{T_\infty^2} \left(k_{hmf} + \frac{16\sigma^* T_\infty^3}{3k^*} \right) \left(\frac{\partial T}{\partial y} \right)^2}_I + \underbrace{\frac{\mu_{nf}}{T_\infty} \left(1 + \frac{1}{\beta} \right) \left(\frac{\partial u}{\partial y} \right)^2}_{II} + \underbrace{\frac{\sigma_{enf}}{T_\infty} \{B_0 u - E_0\}^2}_{III} + \underbrace{\frac{\mu_{nf}}{k' T_\infty} \left(1 + \frac{1}{\beta} \right) u^2}_{IV} \quad (14)$$

- I. Entropy production due to heat transfer
- II. Entropy production due to fluid friction
- III. Entropy production due to Joule heating
- IV. Entropy production due to porous matrix.

By applying equations (8) and (9) in equation (14), the converted equation is

$$N_G = \alpha_1 \left(\frac{k_{hmf}}{k_f} + \frac{4}{3}R \right) \theta'^2 + \frac{\mu_{nf}}{\mu_f} \left(1 + \frac{1}{\beta} \right) Br f'^2 + \frac{\sigma_{enf}}{\sigma_f} M Br f' (f' - E)^2 + \frac{\mu_{nf}}{\mu_f} \left(1 + \frac{1}{\beta} \right) Br K f'^2 \quad (15)$$

where $Br = \frac{\mu_f u_w^2}{k_f \Delta T}$ is the Brinkman number, $\alpha_1 = \frac{\Delta T}{T_\infty}$ is the dimensionless ratio variable and $N_G = \frac{S_G \nu_f T_\infty}{k_f U_0 \Delta T}$ is the local entropy generation.

Bejan number (Be) is as follows:

$$Be = \frac{\text{Heat transfer irreversibility}}{\text{Total entropy generation (Ng)}}$$

$$Be = \frac{\left(\frac{k_{hmf}}{k_f} + \frac{4}{3}R \right) \alpha_1 \theta'^2}{\left(\frac{k_{hmf}}{k_f} + \frac{4}{3}R \right) \alpha_1 \theta'^2 + \frac{\mu_{nf}}{\mu_f} \left(1 + \frac{1}{\beta} \right) Br f'^2 + \frac{\sigma_{enf}}{\sigma_f} M Br f' (f' - E)^2 + \frac{\mu_{nf}}{\mu_f} \left(1 + \frac{1}{\beta} \right) Br K f'^2} \quad (16)$$

Method of solution

Fourth-order Runge–Kutta with shooting method

Higher-order nonlinear differential equations (10) and (11) have been reduced to first-order equations. These are:

$$y'_3 = \left(\frac{\rho_{nf}}{\rho_f} [y_2^2 - y_3 y_1 + F_s y_2^2] + \frac{\sigma_{nf}}{\sigma_f} M y_2 [y_2 - E] + K \frac{\mu_{nf}}{\mu_f} \left(1 + \frac{1}{\beta} \right) y_2 \right) / \frac{\mu_{nf}}{\mu_f} \left(1 + \frac{1}{\beta} \right),$$

$$y'_5 = \left(-\frac{(\rho c_p)_{nf}}{(\rho c_p)_f} y_1 y_5 - Ec \frac{\mu_{nf}}{\mu_f} \left(1 + \frac{1}{\beta} \right) y_3^2 - \frac{\sigma_{nf}}{\sigma_f} M Ec y_2 [y_2 - E]^2 - Q y_4 \right) / \frac{1}{Pr} \left(\frac{k_{nf}}{k_f} + \frac{4}{3}R \right),$$

with boundary conditions.

$$\left. \begin{aligned} y_2(0) = 1, \Pr \frac{\rho_{nf}}{\rho_f} y_1(0) + \frac{k_{nf}}{k_f} Me y_5(0) = 0, \quad y_4(0) = 0, \\ y_2(\infty) = 0, \quad y_4(\infty) = 1. \end{aligned} \right\}$$

where $f = y_1, f' = y_2, f'' = y_3, \theta = y_4$ and $\theta' = y_5$.

The error tolerance is 10^{-8} , and the fourth-order Runge–Kutta (RK) with shooting technique algorithm has a step size of 0.001 (see Figure 2).

Homotopy perturbation method

We applied the homotopy perturbation method (HPM) to the system of nonlinear ordinary differential equations (10) and (11), which are transformed as:

$$(1-p) \left[\begin{aligned} &\frac{\mu_{nf}}{\mu_f} \left(1 + \frac{1}{\beta} \right) f'''(\eta) \\ &-\frac{\mu_{nf}}{\mu_f} \left(1 + \frac{1}{\beta} \right) f_0'''(\eta) \end{aligned} \right] + p \left(\begin{aligned} &\frac{\mu_{nf}}{\mu_f} \left(1 + \frac{1}{\beta} \right) f'''(\eta) - \frac{\rho_{nf}}{\rho_f} \left[(f'(\eta))^2 - f''(\eta) f(\eta) \right] \\ &+ F_s (f'(\eta))^2 \\ &-\frac{\sigma_{nf}}{\sigma_f} M f'(\eta) [f'(\eta) - E] - K \frac{\mu_{nf}}{\mu_f} \left(1 + \frac{1}{\beta} \right) f'(\eta) \end{aligned} \right) \quad (17)$$

$$(1-p) \left[\begin{aligned} &\frac{1}{Pr} \left(\frac{k_{nf}}{k_f} + \frac{4}{3}R \right) \theta''(\eta) \\ &-\frac{1}{Pr} \left(\frac{k_{nf}}{k_f} + \frac{4}{3}R \right) \theta_0''(\eta) \end{aligned} \right] + p \left(\begin{aligned} &\frac{1}{Pr} \left(\frac{k_{nf}}{k_f} + \frac{4}{3}R \right) \theta''(\eta) + \frac{(\rho c_p)_{nf}}{(\rho c_p)_f} f(\eta) \theta'(\eta) \\ &+ \frac{Ec}{(1-\phi)^{2.5}} \left(1 + \frac{1}{\beta} \right) [f''(\eta)]^2 \\ &+ \frac{\sigma_{nf}}{\sigma_f} M Ec f'(\eta) [f'(\eta) - E]^2 + Q \theta(\eta) \end{aligned} \right) \quad (18)$$

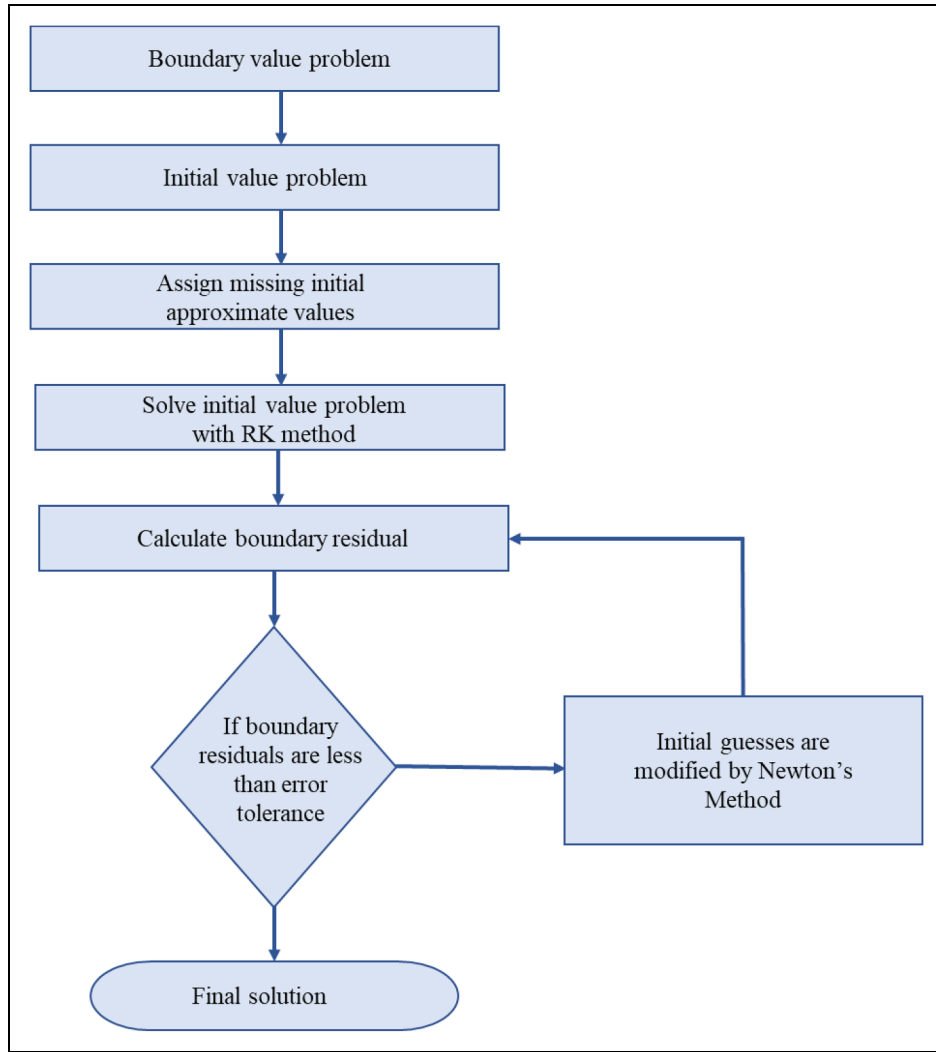


Figure 2. Flow chart of Runge–Kutta (RK) method with shooting method.

The appropriate boundary conditions are as:

$$\left. \begin{aligned} f'(0) &= 1, \text{Pr} \frac{\rho_{nf}}{\rho_f} f(0) + \frac{k_{nf}}{k_f} Me \theta'(0) = 0, \theta(0) = 0, \\ f'(1) &= 0, \theta(1) = 1. \end{aligned} \right\} \quad (19)$$

We ponder $f(\eta)$ and $\theta(\eta)$ as follows:

$$\begin{aligned} f(\eta) &= f_0(\eta) + p f_1(\eta) + p^2 f_2(\eta) + p^3 f_3(\eta) + p^4 f_4(\eta) + \dots, \\ \theta(\eta) &= \theta_0(\eta) + p \theta_1(\eta) + p^2 \theta_2(\eta) + p^3 \theta_3(\eta) + p^4 \theta_4(\eta) + \dots. \end{aligned} \quad (20)$$

Assuming $f_0'''(\eta) = \theta_0''(\eta) = 0$ and substituting $f(\eta)$ and $\theta(\eta)$ in equations (17) and (18), rearranging based on powers of p -terms and then solving the linear differential equations with the corresponding boundary conditions, we get

$$f_0(\eta) = \eta - \frac{1}{2} \eta^2 \quad (21)$$

$$\begin{aligned} f_1(\eta) &= \frac{(8EMS1 - 8FsR1 - 8KM1 - 6MS1 - 9R1)\eta^2}{48M1} \\ &+ \frac{(-EMS1 + FsR1 + KM1 + MS1 + R1)\eta^3}{6M1} \\ &+ \frac{(EMS1 - FsR1 - KM1 - 2MS1 - R1)\eta^4}{24M1} \\ &+ \frac{(2MS1 + R1)\eta^5}{120M1} \\ &\vdots \end{aligned} \quad (22)$$

$$\theta_0(\eta) = \eta, \quad (23)$$

$$\text{where } M1 = \frac{\mu_{nf}}{\mu_f} \left(1 + \frac{1}{\beta}\right), R1 = \frac{\rho_{nf}}{\rho_f}, kn = \frac{k_{nf}}{k_f}, RC1 = \frac{(\rho c_p)_{nf}}{(\rho c_p)_f}, S1 = \frac{\sigma_{nf}}{\sigma_f}, K1 = \frac{1}{\text{Pr}} \left(kn + \frac{4}{3}R\right) \quad (25)$$

Substituting $K = 2$, $E = 2$, $Q = 0.5$, $F_s = 0.5$, $\beta = 1$, $Me = 0$, $Ec = 0.2$, $M = 1.0$, $R = 8$, $Pr = 21$ in equations (21)–(24) and neglecting $O(p^4)$, we get

$$f_0(\eta) = \eta - 0.5\eta^2$$

$$f_1(\eta) = 0.0091\eta^5 - 0.1246\eta^4 + 0.4434\eta^3 - 0.4389\eta^2,$$

$$f_2(\eta) = -0.0001\eta^8 + 0.0022\eta^7 - 0.0193\eta^6 + 0.0726\eta^5 \\ - 0.1093\eta^4 + 0.0878\eta^2,$$

$$f_3(\eta) = 0.0004\eta^9 - 0.0028\eta^8 + 0.0101\eta^7 - 0.0169\eta^6 \\ + 0.0038\eta^5 + 0.0219\eta^4 - 0.0282\eta^2,$$

$$\theta_0(\eta) = \eta,$$

$$\theta_1(\eta) = 0.0021\eta^5 + 0.1999\eta^4 - 1.1655\eta^3 - 0.1476\eta^2 \\ + 1.1111\eta,$$

$$\theta_2(\eta) = 0.0003\eta^8 + 0.0422\eta^7 - 0.3849\eta^6 + 0.7941\eta^5 \\ + 0.5053\eta^4 - 1.1820\eta^3 - 0.1118\eta^2 + 0.3367\eta,$$

$$\theta_3(\eta) = 0.0069\eta^{10} - 0.844\eta^9 + 0.3008\eta^8 - 0.1687\eta^7 \\ - 0.8049\eta^6 + 0.8440\eta^5 + 0.1963\eta^4 \\ - 0.2933\eta^3 - 0.0267\eta^2 + 0.0299\eta,$$

$p = 1$ is the best approximation solution of equation (20); we get:

$$f(\eta) = \eta - 0.8689\eta^2 + 0.4434\eta^3 - 0.2191\eta^4 + 0.0838\eta^5 \\ - 0.0329\eta^6 + 0.0132\eta^7 - 0.0054\eta^8 \\ + 0.0018\eta^9 - 0.0004\eta^{10} + 0.0001\eta^{11} + O(\eta^{12}), \\ \theta(\eta) = 2.4787\eta - 0.2736\eta^2 - 2.6955\eta^3 + 0.9472\eta^4 \\ + 1.8838\eta^5 - 1.6030\eta^6 - 0.3184\eta^7 + 0.8949\eta^8 \\ - 0.2888\eta^9 - 0.0779\eta^{10} + 0.655\eta^{11} - 0.0137\eta^{12} \\ + 0.0009\eta^{13} + O(\eta^{14}).$$

Results and discussion

To obtain the solution of the current model, the HPM (semi-analytical simulation) has been used. The outcomes of essential parameters like melting parameter (Me), Prandtl number (Pr), porosity parameter (K), inertia coefficient (F_s), Eckert number (Ec), radiation parameter (R), electric field parameter (E), heat source parameter (Q), and magnetic parameter (M) on the velocity, temperature, total irreversibility generation, Bejan number, surface friction factor, and rate of heat transfer, are discussed graphically. The values of specific heat (C_p), electrical conductivity (σ), thermal conductivity (k), and density (ρ) of base fluids and nanoparticles are given in Table 1.

Code validation for the HPM and fourth-order RK with shooting method

Figures 3 and 4 show the comparison of the present HPM results in terms of the velocity and temperature graphs with the results of fourth-order RK with shooting method (NM). The comparison indicates a good agreement between semi-analytic method and numerical method results. The predicted results of $-\theta'(0)$ are in nearly full agreement with benchmarking literature as shown in Table 2.

Influences of various parameters on velocity

Figure 5 is plotted to represent the SWCNT-blood nanofluid velocity changes with the growing values of electric field parameter (E) for two distinct melting (Me) cases. It is identified from this figure that the velocity rises with the increasing values of E . The variation on the velocity profile by changing the K is displayed in Figure 6 for numerous values of melting parameter (Me). From Figure 6, it is clear that the rising permeability of medium diminishes the blood nanofluid velocity. Supplemental Figure 7 demonstrates the influence of magnetic parameter (M) on the velocity profile for different combinations of melting parameter (Me). As identified from this figure, the blood velocity rises with the growing values of M since variable electrical conductivity highly influences the fluid motion.

Impacts of numerous parameters on temperature

The effect of nanoparticle volume fraction (ϕ) on the temperature profile for different combinations of melting parameter (Me) are demonstrated in Supplemental Figure 8. From Supplemental Figure 8, we detected that the temperature of the fluid diminishes with higher values of ϕ for different values of Me . Supplemental Figure 9 shows the effect of temperature profile on the dissimilar values of heat source parameter (Q) for different combinations of melting parameter (Me). From Supplemental Figure 9, it is exposed that the temperature profile increase, with the rise in Q . When the external heat is considered in the system, it promotes the average kinetic energy; thus, the molecules move at high speed. Therefore, the blood temperature is rising. Supplemental Figure 10 shows the impact on the temperature profiles for the radiation parameter (R) with different

Table 1. Specific heat (C_p), electrical conductivity (σ), thermal conductivity (k), and density (ρ) of base fluids and CNT.

Physical properties	Blood	SWCNT
$C_p \left(\frac{J}{kg K} \right)$	3594	425
$\sigma \left(\frac{S}{m} \right)$	1.09	4.8×10^{-7}
$k \left(\frac{W}{m K} \right)$	0.492	6600
$\rho \left(\frac{kg}{m^3} \right)$	1063	2600

combinations of melting parameter (Me). It is noticed that the augmenting values of R tend to diminish the blood temperature. Supplemental Figure 11 exemplifies the Eckert

number (Ec) on temperature profile for different combinations of melting parameter (Me). It can be inferred that the temperature enhances with an increase in the Ec .

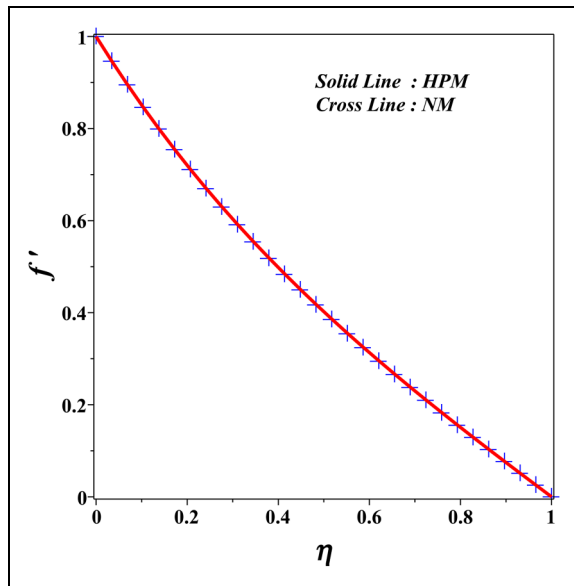


Figure 3. Comparison of the present homotopy perturbation method results in terms of the velocity profile with the results of numeric method.

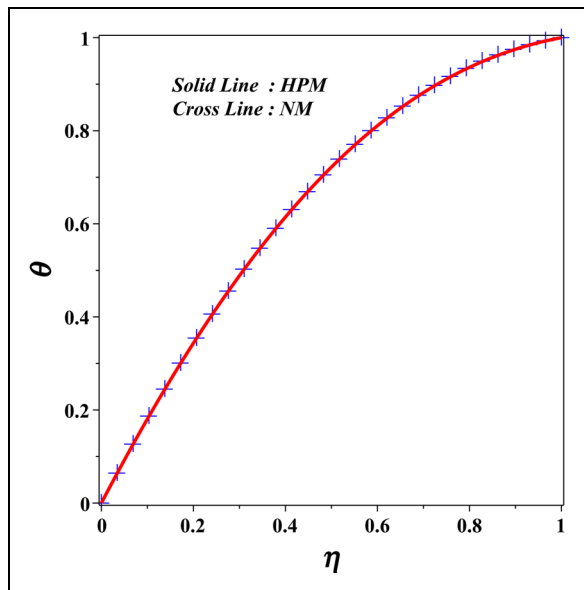


Figure 4. Comparison of HPM with NM on temperature profile.

Impacts of numerous parameters on entropy generation and Bejan number

The impact of magnetic parameter (M) on total irreversibility (entropy) generation and Bejan number is depicted via Supplemental Figures 12 and 13. It is revealed that the higher M promotes the total entropy generation and decays the Bejan number. The entropy generation rate and Bejan number generated due to electric field parameter (E) for different combinations of melting parameter (Me) are illustrated in Supplemental Figures 14 and 15. From these figures it is remarked that the entropy generation rate is enhanced with the increase values of E , the reverse tendency is found for the case of Bejan number. Supplemental Figures 16 and 17 show the nanoparticle volume fraction (ϕ) on the entropy generation rate and Bejan number with different combinations of melting parameter (Me). From these figures, we found that the Bejan number and the entropy generation rate increase with higher values of ϕ . Supplemental Figures 18 and 19 show the impact on the entropy generation rate and Bejan number for Brinkman number (Br) for different combinations of melting parameter (Me). In general, the dimensionless Br is classified as a thermal process to the viscous flowing fluid from the origin. It seems that the larger estimation of Br leads to augments the entropy generation. Because the conduction rate slowly promotes due to dissipation as Br rises. This is the reason for the inflate of entropy generation. Since the entropy rate increases for the larger Br , the Bejan number degrades with an increase in Br . The behavior of radiation parameter (R) for the entropy generation rate and Bejan number are revealed through Supplemental Figures 20 and 21. From these figures, we found that the Bejan number and the entropy generation rate increases with higher values of R .

Effects of various parameters on skin friction and rate of heat transfer

The effect of E , M , F_s and Me on skin friction coefficient is shown in Supplemental Figures 22–24. From this graph, the skin friction coefficient is enriched with high values of E and Me , whereas the opposite behavior is observed in F_s and M . Supplemental Figures 25 and 26 depict the changes in heat transfer rate for varying ϕ , R , and Me . Here, we can observe that the heat transfer rate enhances function of the

Table 2. Comparison of $-\theta'(0)$ for various values of Pr when $\beta = \infty$, $F_s = 0$, $M = 0$, $K = 0$, $R = 0$, $Ec = 0$, $Q = 0$, $\phi = 0$, and $Me = 0$ with previous work.^{21,23}

Pr	0.72	1.0	3.0	7.0	10.0	100.0
Ishak et al. ²¹ (Exact Sol.)		1.0000	1.9237		3.7207	
Pal et al. ²³	0.80863135	1.0000	1.92368256	3.07225020	3.72067391	12.2940835
Present results	0.808631	1.0000	1.923683	3.072250	3.720674	12.294083

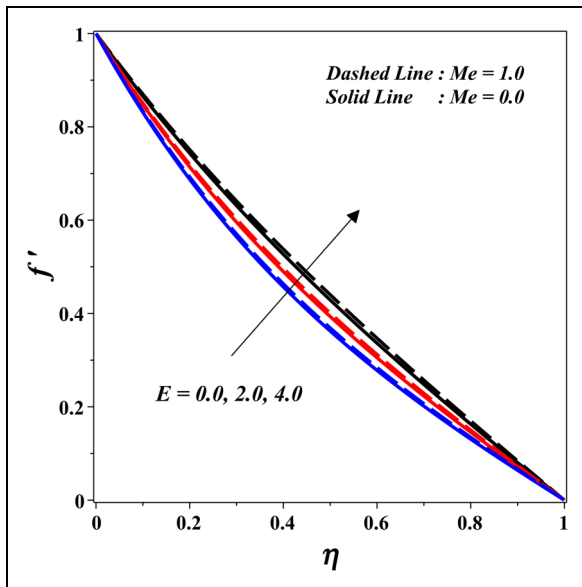


Figure 5. Velocity profile for electric field parameter.

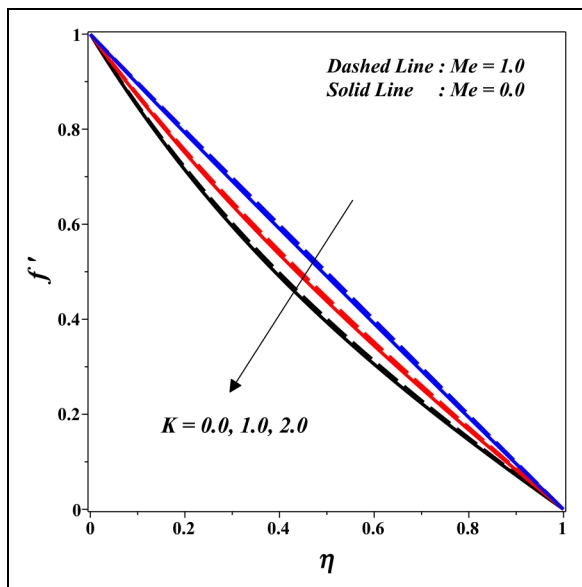


Figure 6. Velocity profile for porosity parameter.

higher values of Me . And also, it is clear that for higher values of ϕ and R cause lower in the heat transfer rate.

Conclusion

Current paper analyzed the study on electro-magnetohydrodynamic SWCNT-blood nanofluid flow with the impacts of electro-conductive heating, ohmic dissipation, and thermal radiation. Melting/non-melting heat transfer is used in biomedical applications such as, to destroy tissues and cells, biopreservation, an innovative technology applied to tissue banking and cell banking, plastic surgery treatment, ART, cell therapeutics, tissue engineering, and organ transplantation. The salient observations of this investigation are as follows:

- The low radiation frequency used in this FDA-destroyed technology raises the temperature, causing the cells to degenerate and die. The surface naturally repairs the treated area and removes dead fat cells permanently.
- The velocity of the blood increases with higher values of electric field parameter causing the friction of the cells increase.
- The velocity of the blood increases with increasing the inertia coefficient.
- An enhancement in entropy generation is observed with the rising values of the electric field parameter.

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Supplemental material

Supplemental material for this article is available online.

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